

# **FYBCA Semester 2 — Complete Exam Notes**

## **Subject 5: Introduction to Graphics Design**

KBCNMU | Exam: 30 Marks | Complete Notes

# TOPIC 1: Basics of Graphic Design

## 1.1 What is Graphic Design?

Graphic Design is the art and practice of creating visual content to communicate information and ideas. It combines images, text, colors, and layout to solve visual communication problems. Used in advertising, branding, websites, print media, and packaging.

## 1.2 Elements of Graphic Design

| Element    | Description                                                                  |
|------------|------------------------------------------------------------------------------|
| Line       | Basic element — straight, curved, diagonal. Creates structure and direction. |
| Shape      | Two-dimensional area: geometric (circle, square) or organic (freeform).      |
| Color      | Creates mood and emotion. Described using hue, saturation, brightness.       |
| Texture    | Surface quality — rough, smooth, soft. Adds depth and interest.              |
| Space      | Positive space (subject) and Negative space (background).                    |
| Form       | 3D representation of shapes: cube, sphere, cylinder.                         |
| Typography | Art of arranging text: font, size, spacing, alignment.                       |
| Value      | Lightness or darkness of a color.                                            |

## TOPIC 2: Principles of Design

### 2.1 The 8 Principles of Design

| Principle       | Definition                                                               | Example                             |
|-----------------|--------------------------------------------------------------------------|-------------------------------------|
| Balance         | Equal distribution of visual weight. Can be symmetrical or asymmetrical. | Centered logo (symmetrical)         |
| Contrast        | Use of opposites to create visual interest: light/dark, big/small.       | Black text on white background      |
| Emphasis        | Making one element stand out as the focal point.                         | Bold headline text                  |
| Proportion      | Size relationship between elements.                                      | Large title, small body text        |
| Rhythm/Movement | Repeated elements that guide the eye through the design.                 | Repeated patterns, borders          |
| Variety         | Using different elements to create visual interest and avoid monotony.   | Different fonts, colors, shapes     |
| Unity           | All elements work together as a cohesive whole.                          | Consistent colors, fonts throughout |
| Alignment       | Elements lined up to create order and organization.                      | Text aligned to left margin         |

#### Important Points

- Balance: Symmetrical (equal sides) vs Asymmetrical (unequal but visually balanced).
- Contrast is essential for readability — never use low-contrast text on background.
- Rule of thirds: Divide design into 9 equal parts; place key elements at intersections.
- Negative space is as important as positive space.

## TOPIC 3: Color Models

### 3.1 RGB Color Model

RGB (Red, Green, Blue) is an ADDITIVE color model used for SCREENS (monitors, TVs, phones). Colors are created by adding light together.

- Red + Green + Blue = White (all colors at full intensity)
- No color = Black (all values at 0)
- Each channel: 0-255 (or 0%-100%)
- RGB(255,0,0) = Red; RGB(0,255,0) = Green; RGB(0,0,255) = Blue
- Used in: websites (CSS), digital displays, Photoshop screen design

### 3.2 CMYK Color Model

CMYK (Cyan, Magenta, Yellow, Key/Black) is a SUBTRACTIVE color model used for PRINTING. Colors are created by subtracting light.

- Used by printers using ink: Cyan, Magenta, Yellow, and Black inks
- K = Key = Black (added for deeper blacks)
- Values expressed as percentage (0-100%)
- Cyan + Magenta + Yellow = Black (in theory, but actual result is muddy brown — so K added)
- Used in: newspapers, magazines, business cards, brochures

### 3.3 HSB/HSV Color Model

HSB (Hue, Saturation, Brightness) — also called HSV (Hue, Saturation, Value):

| Component        | Description                                    | Range                                |
|------------------|------------------------------------------------|--------------------------------------|
| Hue              | The actual color (red, blue, green, yellow...) | 0° to 360° on color wheel            |
| Saturation       | Intensity/purity of color                      | 0% (gray) to 100% (vivid)            |
| Brightness/Value | Lightness of color                             | 0% (black) to 100% (full brightness) |

### 3.4 Comparison of Color Models

| Feature        | RGB              | CMYK                         | HSB                         |
|----------------|------------------|------------------------------|-----------------------------|
| Use            | Screens/digital  | Printing                     | Design software             |
| Primary Colors | Red, Green, Blue | Cyan, Magenta, Yellow, Black | Hue, Saturation, Brightness |
| Type           | Additive         | Subtractive                  | Descriptive                 |
| Values         | 0-255 each       | 0-100% each                  | 0-360, 0-100%, 0-100%       |

# TOPIC 4: Adobe Photoshop Tools

## 4.1 What is Photoshop?

Adobe Photoshop is the industry-standard software for image editing, photo retouching, digital painting, and graphic design. It works with RASTER (pixel-based) images.

## 4.2 Photoshop Toolbox

| Tool                | Shortcut | Description                                        |
|---------------------|----------|----------------------------------------------------|
| Move Tool           | V        | Moves layers and selections around the canvas.     |
| Rectangular Marquee | M        | Selects rectangular or square areas.               |
| Elliptical Marquee  | M        | Selects circular/elliptical areas.                 |
| Lasso Tool          | L        | Freehand selection by drawing.                     |
| Magic Wand          | W        | Selects areas of similar color automatically.      |
| Quick Selection     | W        | Brushes over area to select similar pixels.        |
| Crop Tool           | C        | Crops/trims image to desired size.                 |
| Eyedropper          | I        | Samples color from image.                          |
| Healing Brush       | J        | Removes blemishes, repairs imperfections.          |
| Clone Stamp         | S        | Paints one area using sampled pixels from another. |
| Eraser Tool         | E        | Erases pixels to background color or transparency. |
| Gradient Tool       | G        | Creates gradual blend between colors.              |
| Paint Bucket        | G        | Fills area with foreground color.                  |
| Brush Tool          | B        | Paints strokes with chosen color and style.        |
| Pencil Tool         | B        | Draws hard-edged strokes.                          |
| Type Tool           | T        | Adds text to image (horizontal or vertical).       |
| Pen Tool            | P        | Creates precise paths/shapes.                      |
| Zoom Tool           | Z        | Zooms in and out of image.                         |
| Hand Tool           | H        | Pans/scrolls around a zoomed image.                |

## 4.3 Key Photoshop Concepts

| Concept    | Description                                                                                   |
|------------|-----------------------------------------------------------------------------------------------|
| Layers     | Like transparent sheets stacked on top of each other. Each layer can be edited independently. |
| Layer Mask | Non-destructive way to hide parts of a layer without deleting pixels.                         |

|                   |                                                                                  |
|-------------------|----------------------------------------------------------------------------------|
| Adjustment Layers | Allows color/tone adjustments without changing original pixels.                  |
| Selection         | An area defined for editing. Tools: Marquee, Lasso, Magic Wand, Quick Selection. |
| Filters           | Pre-set effects: blur, sharpen, distort, texture, artistic effects.              |
| History Panel     | Records all edits — allows undo to any previous state.                           |
| Blending Modes    | Determines how layers interact with layers below (Multiply, Screen, Overlay).    |
| Resolution        | Pixels per inch (PPI). 72 ppi for web, 300 ppi for print.                        |

# TOPIC 5: CorelDRAW Tools

## 5.1 What is CorelDRAW?

CorelDRAW is a professional VECTOR GRAPHICS software used for illustration, logo design, layout, and page design. It works with VECTOR (mathematical path-based) images that can scale to any size without losing quality.

## 5.2 Raster vs Vector

| Feature      | Raster (Photoshop)                | Vector (CorelDRAW)                      |
|--------------|-----------------------------------|-----------------------------------------|
| Made of      | Pixels (dots)                     | Mathematical paths and curves           |
| Scalability  | Loses quality when scaled up      | Scales to any size without quality loss |
| File size    | Larger                            | Smaller (usually)                       |
| Best for     | Photos, textures, detailed images | Logos, icons, illustrations             |
| File formats | JPEG, PNG, BMP, PSD               | SVG, AI, CDR, EPS                       |

## 5.3 CorelDRAW Toolbox

| Tool                | Description                                        |
|---------------------|----------------------------------------------------|
| Pick Tool           | Selects, moves, resizes, and rotates objects.      |
| Shape Tool          | Edits individual nodes (points) of objects.        |
| Rectangle Tool      | Draws rectangles and squares.                      |
| Ellipse Tool        | Draws circles and ellipses.                        |
| Polygon Tool        | Draws polygons and stars.                          |
| Text Tool           | Adds and edits text (Artistic or Paragraph).       |
| Pen Tool            | Creates freehand Bezier curves and paths.          |
| Freehand Tool       | Draws freehand lines and curves.                   |
| Fill Tool           | Applies fill: uniform, gradient, pattern, texture. |
| Outline Tool        | Controls outline/border color and thickness.       |
| Zoom Tool           | Zooms in/out of the drawing area.                  |
| Hand Tool           | Pans around the drawing area.                      |
| Eyedropper          | Samples color from existing objects.               |
| Interactive Effects | Drop shadow, transparency, blend, extrude tools.   |

## 5.4 CorelDRAW Key Features

- Node Editing: Edit individual anchor points to reshape objects precisely.
- Bezier Curves: Create smooth curves by manipulating control handles.

- Layers: Organize design elements on separate layers.
- Master Page: Common elements visible on all pages.
- Envelope Tool: Distorts objects into envelope shapes.
- Blend Tool: Creates gradual transition between two objects.
- Contour Tool: Creates concentric copies of an object.



# TOPIC 6: Image Editing Concepts

## 6.1 Cropping and Resizing

CROPPING removes unwanted parts of an image to focus on subject. RESIZING changes the physical dimensions of an image.

| Action       | Description                                                             |
|--------------|-------------------------------------------------------------------------|
| Crop         | Remove edges of image to improve composition or remove unwanted areas.  |
| Resize       | Change width/height of image. Can maintain or break aspect ratio.       |
| Aspect Ratio | Proportional relationship of width to height (e.g., 16:9, 4:3).         |
| Resampling   | Adding or removing pixels when resizing (Bicubic, Bilinear algorithms). |

## 6.2 Color Correction and Adjustments

| Adjustment    | Description                                    |
|---------------|------------------------------------------------|
| Brightness    | Overall lightness/darkness of image.           |
| Contrast      | Difference between lightest and darkest areas. |
| Saturation    | Intensity/vividness of colors.                 |
| Hue           | Shifts all colors in the image.                |
| Levels        | Adjusts shadow, midtone, highlight separately. |
| Curves        | Fine-tuned control over tone and color.        |
| Color Balance | Adjusts balance between complementary colors.  |

## 6.3 Filters and Effects

| Category | Examples                                               |
|----------|--------------------------------------------------------|
| Blur     | Gaussian Blur, Motion Blur, Lens Blur — softens image. |
| Sharpen  | Unsharp Mask, Smart Sharpen — enhances edges.          |
| Distort  | Wave, Ripple, Twirl, Spherize — warps image.           |
| Stylize  | Emboss, Find Edges, Glowing Edges — artistic effects.  |
| Texture  | Grain, Mosaic Tiles, Stained Glass.                    |
| Artistic | Watercolor, Oil Paint, Sketch effects.                 |

## TOPIC 7: File Formats

| Format   | Full Name                        | Type                 | Best Used For                                  |
|----------|----------------------------------|----------------------|------------------------------------------------|
| JPEG/JPG | Joint Photographic Experts Group | Raster, Lossy        | Photos, web images. Small file size.           |
| PNG      | Portable Network Graphics        | Raster, Lossless     | Images with transparency. Web logos.           |
| GIF      | Graphics Interchange Format      | Raster, Lossless     | Animations, simple graphics. 256 colors only.  |
| BMP      | Bitmap                           | Raster, Uncompressed | Windows images. Very large file size.          |
| TIFF     | Tagged Image File Format         | Raster, Lossless     | Professional printing. High quality.           |
| PSD      | Photoshop Document               | Raster, Native       | Photoshop working files with layers.           |
| SVG      | Scalable Vector Graphics         | Vector               | Web icons, logos. Scales without quality loss. |
| AI       | Adobe Illustrator                | Vector               | Illustrator working files.                     |
| CDR      | CorelDRAW Document               | Vector               | CorelDRAW working files.                       |
| EPS      | Encapsulated PostScript          | Vector/Raster        | Professional printing and publishing.          |
| PDF      | Portable Document Format         | Both                 | Universal sharing, printing.                   |

### Lossy vs Lossless Compression

| Feature    | Lossy                         | Lossless                    |
|------------|-------------------------------|-----------------------------|
| Definition | Some data permanently removed | All original data preserved |
| File Size  | Much smaller                  | Larger                      |
| Quality    | Degrades with each save       | No quality loss             |
| Examples   | JPEG, MP3                     | PNG, BMP, TIFF              |

## TOPIC 8: Animation Concepts

### 8.1 What is Animation?

Animation is the technique of creating the illusion of motion by displaying a sequence of still images (frames) rapidly. Based on PERSISTENCE OF VISION — human eye retains an image for a fraction of a second.

### 8.2 Types of Animation

| Type                       | Description                                                   | Examples                       |
|----------------------------|---------------------------------------------------------------|--------------------------------|
| Traditional/Cell Animation | Hand-drawn frames on transparent celluloid sheets.            | Old Disney movies, Tom & Jerry |
| Computer Animation (2D)    | Using software to create animated scenes in 2D.               | Adobe Animate, Toon Boom       |
| Computer Animation (3D)    | Creating three-dimensional animated objects and environments. | Pixar movies, game graphics    |
| Stop Motion                | Physical objects photographed frame by frame.                 | Claymation, Wallace & Gromit   |
| GIF Animation              | Looping animated images in GIF format.                        | Website loading icons, memes   |
| CSS Animation              | Web animations using CSS transitions and keyframes.           | Button hover effects, loaders  |

### 8.3 Key Animation Terms

| Term                    | Definition                                                       |
|-------------------------|------------------------------------------------------------------|
| Frame                   | A single still image in an animation sequence.                   |
| FPS (Frames Per Second) | Speed at which frames are displayed. Film: 24fps, Games: 60fps.  |
| Keyframe                | Important frame defining start/end of a transition.              |
| Tweening                | Automatic generation of intermediate frames between keyframes.   |
| Storyboard              | Visual planning tool showing sequence of frames/scenes.          |
| Timeline                | Visual representation of animation sequence in editing software. |
| Morphing                | Smooth transformation of one image/shape into another.           |

### 8.4 Applications of Graphic Design

| Industry                | Applications                                        |
|-------------------------|-----------------------------------------------------|
| Advertising & Marketing | Banners, posters, social media graphics, brochures. |

|             |                                                             |
|-------------|-------------------------------------------------------------|
| Web Design  | UI/UX design, website layouts, icons, banners.              |
| Print Media | Newspapers, magazines, books, packaging.                    |
| Film & TV   | Title sequences, special effects, credits.                  |
| Education   | Textbook illustrations, e-learning materials, infographics. |
| Corporate   | Business cards, letterheads, presentations, reports.        |
| Gaming      | Game art, character design, environment design.             |

### Must Remember

- RGB: For screens (Red+Green+Blue = White). Additive model.
- CMYK: For printing (Cyan+Magenta+Yellow+Black). Subtractive model.
- Raster (Photoshop): Pixel-based. Loses quality when scaled up.
- Vector (CorelDRAW): Path-based. Scales infinitely without quality loss.
- JPEG: Lossy compression — good for photos.
- PNG: Lossless with transparency — good for logos on web.
- Resolution: 72 PPI for screen; 300 PPI for print quality.
- FPS: Films use 24fps; video games typically 60fps.

### Exam Questions

- Explain the principles of design with examples. (6 marks)
- What are color models? Differentiate between RGB and CMYK. (6 marks)
- Explain important tools of Photoshop. (6 marks)
- Differentiate between raster and vector graphics. (4 marks)
- What are different file formats used in graphic design? (4 marks)
- What is animation? Explain its types. (6 marks)

## QUICK REVISION SUMMARY — Subject 5: Graphics Design

| Topic               | Key Points                                                                                  |
|---------------------|---------------------------------------------------------------------------------------------|
| Graphic Design      | Visual communication using images, text, colors, layout.                                    |
| Design Elements     | Line, Shape, Color, Texture, Space, Form, Typography, Value.                                |
| Design Principles   | Balance, Contrast, Emphasis, Proportion, Rhythm, Unity, Alignment, Variety.                 |
| RGB                 | Additive color model for SCREENS. R+G+B = White. Values 0-255.                              |
| CMYK                | Subtractive color model for PRINTING. C+M+Y+K=Black. Values 0-100%.                         |
| Raster              | Pixel-based (Photoshop). Loses quality when scaled. JPEG, PNG.                              |
| Vector              | Path-based (CorelDRAW). Scales without quality loss. SVG, AI, CDR.                          |
| Photoshop Key Tools | Move(V), Crop(C), Brush(B), Type(T), Layers, Selection tools.                               |
| CorelDRAW Key Tools | Pick tool, Shape tool, Pen, Blend, Contour, Envelope.                                       |
| File Formats        | JPEG(photos), PNG(transparency), GIF(animation), PSD(Photoshop), CDR(CorelDRAW).            |
| Animation           | Rapid display of frames creates illusion of motion. Types: Traditional, 2D/3D, Stop-motion. |
| FPS                 | Frames Per Second. Film=24fps, Gaming=60fps.                                                |